

# VEDANT CHANDRA

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## Research Interests

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- **The Milky Way:** dark matter, galactic archaeology, spectroscopic surveys, galaxy formation, stellar streams, dwarf galaxies, hydrodynamical simulations, chemical evolution, near-field cosmology;
- **Optical Instrumentation:** multi-object spectroscopy, robotic fiber positioning, metrology, software;
- **Stellar Evolution:** white dwarfs, binary stars, supernovae;

## Professional Appointments

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|------------------------------------------------------------------------------|-----------|
| Postdoctoral Associate, Massachusetts Institute of Technology                | 2026 –    |
| Graduate Research Assistant, Center for Astrophysics   Harvard & Smithsonian | 2021–2026 |
| Visiting Researcher, Max Planck Institute for Astronomy                      | 2022–2026 |
| Undergraduate Research Intern, Space Telescope Science Institute             | 2020–2021 |
| Undergraduate Research Assistant, Johns Hopkins University & Hospital        | 2018–2021 |

## Education

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| Harvard University                                                                                                                                                                                                | 2021–2026 |
| <ul style="list-style-type: none"><li>• A.M., Ph.D. Astronomy &amp; Astrophysics</li><li>• “Mapping the Uncharted Horizons of the Milky Way” (<a href="#">dissertation</a>, <a href="#">colloquium</a>)</li></ul> |           |
| Johns Hopkins University                                                                                                                                                                                          | 2017–2021 |
| <ul style="list-style-type: none"><li>• B.S. Physics &amp; Applied Mathematics, minor in Space Sciences</li></ul>                                                                                                 |           |

## Awards & Honors

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|----------------------------------------------------------------------------------------------------------------|------|
| <a href="#">Fireman Prize</a> , Harvard University                                                             | 2026 |
| <i>for the most outstanding PhD thesis in the field of astrophysics</i>                                        |      |
| <a href="#">Hubble Fellowship</a> , NASA                                                                       | 2026 |
| <i>for outstanding postdoctoral scientists to pursue independent research in any area of NASA Astrophysics</i> |      |
| <a href="#">School of Science Postdoctoral Fellowship</a> , MIT                                                | 2026 |
| <i>for scientific excellence, innovation, and cross-disciplinary research</i>                                  |      |
| <a href="#">CCAPP Price Prize in Cosmology and Astrophysics</a>                                                | 2025 |
| <i>to two PhD students worldwide for research excellence and exceptional promise</i>                           |      |
| <a href="#">Peirce Fellowship</a> , Harvard University                                                         | 2021 |
| <i>to incoming PhD students who possess significant promise as researchers</i>                                 |      |
| <a href="#">Chambliss Astronomy Achievement Award</a> , American Astronomical Society                          | 2021 |
| <i>to recognize exemplary research by an undergraduate student</i>                                             |      |
| <a href="#">Dean’s and Provost’s Undergraduate Research Awards</a> , JHU                                       | 2019 |
| <i>to recognize and support exceptional research by undergraduate students</i>                                 |      |

## Publication Summary

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65 refereed publications, 11 as lead author, 14 as second/supervising author;  $h$ -index = 26 ([ADS Library](#))

## Research Mentorship

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|                                                                                             |              |
|---------------------------------------------------------------------------------------------|--------------|
| <a href="#">Haniyeh Tajer</a> (Ohio State PhD, stellar streams in SDSS-V)                   | 2026–Present |
| <a href="#">Hongbo Xia</a> (Northwestern UG, automated observing at the MMT)                | 2026–Present |
| <a href="#">Astrid Liu</a> (Harvard UG, modeling stellar streams in variable MW potentials) | 2026–Present |
| <a href="#">Catherine Miller</a> (Harvard PhD, the Via robotic fiber positioner)            | 2025–Present |
| <a href="#">Camille Chiu</a> (Yale UG → Harvard PhD, dynamics of the Milky Way)             | 2025–Present |
| <a href="#">Abigail White</a> (Harvard PhD, fiber optics for Via)                           | 2024–Present |
| <a href="#">Gautham A. Pallathadka</a> (JHU PhD, WD-WD binaries in SDSS-V, 3 papers)        | 2022–2025    |
| <a href="#">Stefan Arseneau</a> (JHU UG → BU PhD, the WD mass-radius relation, 1 paper)     | 2022–2025    |
| <a href="#">Nicole Crumpler</a> (JHU PhD, WDs in SDSS-V, 2 papers)                          | 2022–2024    |

## Grant Allocations as PI

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|--------------------------------------------------------------------|-----------|
| STScI JWST Discretionary Fund (\$42,740, Co-PI with Mario Gennaro) | 2020      |
| JHU Dean's and Provost's Undergraduate Research Awards (\$13,500)  | 2019–2020 |

## Observatory Allocations

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### Principal Investigator:

|                                                    |           |
|----------------------------------------------------|-----------|
| MPG/ESO 2.2m, FEROS, 250 hours                     | 2023–2024 |
| ESO Very Large Telescope, FLAMES/GIRAFFE, 75 hours | 2023      |
| Magellan Observatory, MagE, 19 nights              | 2022–2023 |
| MMT Observatory, Hectochelle, 4 nights             | 2022–2023 |
| Anglo-Australian Telescope, 2dF+AAOmega, 3 nights  | 2022      |
| Gemini Observatory, GMOS, 30 hours                 | 2021–2022 |
| Apache Point Observatory, DIS & ARCTIC, 6 nights   | 2020–2021 |
| Neils Gehrels Swift Observatory, UVOT, 1 hour      | 2021      |

### Co-Investigator:

|                                              |           |
|----------------------------------------------|-----------|
| MMT Observatory, Hectochelle, 5 nights       | 2025      |
| WIYN 3.5m, NEID, 3 nights                    | 2025      |
| ESO Very Large Telescope, X-Shooter, 8 hours | 2023      |
| CTIO Blanco, DECam, 54 nights                | 2023      |
| James Webb Space Telescope, NIRCAM, 48 hours | 2023      |
| Gemini Observatory, GMOS, 7 hours            | 2023      |
| Magellan Observatory, MagE, 7 nights         | 2022–2023 |
| Apache Point Observatory, DIS, 3 nights      | 2022      |

## Teaching

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|                                                               |              |
|---------------------------------------------------------------|--------------|
| Teaching Fellow, ASTRON 202a: Galaxies and Cosmology, Harvard | Spring, 2024 |
| Teaching Fellow, ASTRON 120: Stellar Physics, Harvard         | Spring, 2023 |
| Teaching Assistant, 360.133: Great Books at Hopkins, JHU      | Fall, 2018   |
| Teaching Assistant, 171.101: General Physics I, JHU           | Summer, 2018 |

## Professional Service

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|-------------------------------------------------------------------------|--------------|
| Deputy Project Scientist, <a href="#">Via Project</a>                   | 2022-Present |
| Journal Referee (11×: Nature, ApJ, ApJL, ApJS, MNRAS, A&A, PASP, JATIS) | 2022-Present |
| Faculty Search Committee, Harvard Astronomy                             | 2023-2024    |
| Lead Organizer, Harvard Astronomy Student-Faculty Forum                 | 2023-2024    |
| Representative, Harvard Astronomy Student-Faculty Council               | 2021-2023    |

## Outreach

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|                                                                                                                                                                                |              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| Representative, Astronomy Graduate Student Congress                                                                                                                            | 2023-2026    |
| Guest Speaker                                                                                                                                                                  | 2023-Present |
| <ul style="list-style-type: none"><li>• Gloucester Area Astronomy Club, 50 attendees (<a href="#">video</a>)</li><li>• The Shri Ram School, 200 high-school students</li></ul> |              |
| Executive Committee, CfA Social & Recreational Club                                                                                                                            | 2021-2023    |

## Selected Press Coverage

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|-------------------------------------------------------------------------------------------------------------------------------------------|
| Discovery in SDSS-V of the most metal-poor star yet known ( <a href="#">Ji, Chandra et al 2026</a> ; <a href="#">Chandra et al 2026</a> ) |
| • <a href="#">Carnegie Science</a> , <a href="#">New Scientist</a> , <a href="#">Gizmodo</a> , <a href="#">The Independent</a>            |
| Discovery of the Magellanic Stellar Stream ( <a href="#">Chandra et al 2023</a> )                                                         |
| • <a href="#">CfA Press Release</a> , <a href="#">Sky &amp; Telescope Magazine</a> , <a href="#">New Scientist</a>                        |
| Discovery of the ‘Poor Old Heart’ of the Milky Way ( <a href="#">Rix, Chandra et al 2022</a> )                                            |
| • <a href="#">MPIA Press Release</a> , <a href="#">Quanta Magazine</a> , <a href="#">ScienceNews Magazine</a>                             |
| Novel measurements of the white dwarf mass-radius relation ( <a href="#">Chandra et al 2021</a> )                                         |
| • <a href="#">JHU Press Release</a> , <a href="#">ScienceNews Magazine</a>                                                                |

## Talks

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### Invited Talks and Seminars († indicates invited plenary/review)

|                                                                     |                 |
|---------------------------------------------------------------------|-----------------|
| Galaxy Lunch, Yale University                                       | April, 2026     |
| Astrophysics Seminar, University of Notre Dame                      | March, 2026     |
| KICP Seminar, University of Chicago                                 | October, 2025   |
| Lunch Talk, UC Berkeley                                             | October, 2025   |
| KIPAC Tea Talk, Stanford University                                 | October, 2025   |
| Tea Talk, California Institute of Technology                        | September, 2025 |
| Lunch Talk, Carnegie Observatories                                  | September, 2025 |
| Afternoon Science Talk, MIT Kavli Institute                         | September, 2025 |
| CCAPP Price Prize Seminar, The Ohio State University                | September, 2025 |
| †XMC II: Clouds Over Yellowstone, Bozeman ( <a href="#">video</a> ) | May, 2025       |
| Tea Talk, California Institute of Technology                        | November, 2024  |
| Decade of Illustris Meeting, Castello di Gargonza                   | October, 2024   |
| Wine & Cheese Seminar, Johns Hopkins University                     | September, 2024 |
| Astrophysics Coffee, The Institute for Advanced Study               | September, 2024 |
| Galaxy Coffee, Max Planck Institute for Astronomy                   | June, 2024      |

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|------------------------------------------------------------------------|----------------|
| Galaxy Coffee, Max Planck Institute for Astronomy                      | July, 2023     |
| Online Meetings on Evolved Stars and Systems ( <a href="#">video</a> ) | December, 2021 |
| Astrophysics Coffee, Institute for Advanced Study                      | October, 2021  |

### **Contributed Talks**

|                                                                     |                 |
|---------------------------------------------------------------------|-----------------|
| NHFP Symposium, CfA   Harvard % Smithsonian                         | October, 2026   |
| Probe Combination for DM Physics, The University of Chicago         | September, 2026 |
| MMT Symposium, The University of Arizona                            | April, 2026     |
| AAS 247 Winter Meeting, The American Astronomical Society           | January, 2026   |
| SDSS-V Collaboration Meeting, Max Planck Institute for Astronomy    | June, 2025      |
| Magellan Science Meeting, Carnegie EPL                              | May, 2025       |
| The Milky Way Assembly Tale, INAF Bologna                           | May, 2024       |
| Milky Clouds Above Manhattan, Flatiron CCA                          | February, 2024  |
| ITC Luncheon, CfA   Harvard & Smithsonian ( <a href="#">video</a> ) | February, 2024  |
| Supernova Explosions (SNE <sub>x</sub> ) Meeting                    | November, 2023  |
| Central Halo Workshop, University of Cambridge                      | September, 2023 |
| ITC Luncheon, CfA   Harvard & Smithsonian ( <a href="#">video</a> ) | September, 2023 |
| SDSS-V Collaboration Meeting, Flatiron CCA                          | August, 2023    |
| Gaia XPloration Workshop, University of Cambridge                   | May, 2023       |
| Wide Field Spectroscopy vs Galaxy Formation Theory                  | March, 2023     |
| Disk Formation Workshop, UC Irvine                                  | September, 2022 |
| ITC Luncheon, CfA   Harvard & Smithsonian ( <a href="#">video</a> ) | September, 2022 |
| Space Telescope Science Institute, Summer Symposium                 | July, 2020      |
| Space Telescope Science Institute, Summer Symposium                 | August, 2019    |
| Maryland Space Grant Consortium, Annual Symposium                   | July, 2019      |

### **References**

|                                                         |                                                                                     |
|---------------------------------------------------------|-------------------------------------------------------------------------------------|
| Charlie Conroy, Harvard University                      | PhD Advisor, <a href="mailto:cconroy@cfa.harvard.edu">cconroy@cfa.harvard.edu</a>   |
| Hans-Walter Rix, Max Planck Institute for Astronomy     | Advisor, <a href="mailto:rix@mpia.de">rix@mpia.de</a>                               |
| Daniel Fabricant, Smithsonian Astrophysical Observatory | Advisor, <a href="mailto:dfabricant@cfa.harvard.edu">dfabricant@cfa.harvard.edu</a> |
| Nadia L. Zakamska, Johns Hopkins University             | Advisor, <a href="mailto:zakamska@jhu.edu">zakamska@jhu.edu</a>                     |

## Publication List ([ADS Library](#))

Project advisees are highlighted with an asterisk\*

### Lead-Author Publications

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11. **Vedant Chandra**, Phillip A. Cargile, Alexander P. Ji, et al. (2026)  
“Mapping the Distant and Metal-poor Milky Way with SDSS-V”  
[The Astrophysical Journal, 1000, 283](#)
10. **Vedant Chandra**, Rohan P. Naidu, Charlie Conroy, et al. (2025)  
“All-sky Kinematics of the Distant Halo: The Reflex Response to the LMC”  
[The Astrophysical Journal, 988, 156](#)
9. **Vedant Chandra**, Vadim A. Semenov, Hans-Walter Rix, et al. (2024)  
“The Three-phase Evolution of the Milky Way”  
[The Astrophysical Journal, 972, 112](#)
8. **Vedant Chandra**, Rohan P. Naidu, Charlie Conroy, et al. (2023)  
“Discovery of the Magellanic Stellar Stream Out to 100 kpc”  
[The Astrophysical Journal, 956, 110](#)
7. **Vedant Chandra**, Rohan P. Naidu, Charlie Conroy, et al. (2023)  
“Distant Echoes of the Milky Way’s Last Major Merger”  
[The Astrophysical Journal, 951, 26](#)
6. **Vedant Chandra**, Charlie Conroy, Nelson Caldwell, et al. (2022)  
“A Ghost in Boötes: The Least-Luminous Disrupted Dwarf Galaxy”  
[The Astrophysical Journal, 940, 127](#)
5. **Vedant Chandra**, Hsiang-Chih Hwang, Nadia L. Zakamska, et al. (2022)  
“The SN Ia runaway LP 398-9: detection of circumstellar material and surface rotation”  
[Monthly Notices of the Royal Astronomical Society, 512, 6122](#)
4. **Vedant Chandra**, Hsiang-Chih Hwang, Nadia L. Zakamska, et al. (2021)  
“A 99 minute Double-lined White Dwarf Binary from SDSS-V”  
[The Astrophysical Journal, 921, 160](#)
3. **Vedant Chandra** & Kevin C. Schlaufman (2021)  
“Searching for Low-mass Population III Stars Disguised as White Dwarfs”  
[The Astronomical Journal, 161, 197](#)
2. **Vedant Chandra**, Hsiang-Chih Hwang, Nadia L. Zakamska, Tamás Budavári (2020)  
“Computational tools for the spectroscopic analysis of white dwarfs”  
[Monthly Notices of the Royal Astronomical Society, 497, 2688](#)
1. **Vedant Chandra**, Hsiang-Chih Hwang, Nadia L. Zakamska, Sihao Cheng (2020)  
“A Gravitational Redshift Measurement of the White Dwarf Mass-Radius Relation”  
[The Astrophysical Journal, 899, 146](#)

### Publications as Co-Lead or Supervising Author

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14. Alexander P. Ji, **Vedant Chandra**, Selenna Mejias-Torres, et al. (2026)  
“A nearly pristine star from the Large Magellanic Cloud”  
[Nature Astronomy, 10, 842](#)
13. Jéa Adams Redai, **Vedant Chandra**, Samuel W. Yee, et al. (2026)  
“An Ancient Brown Dwarf Transiting a Metal-poor Thick-disk Star”  
[The Astrophysical Journal Letters, 1002, L41](#)

12. Gautham A. Pallathadka\*, **Vedant Chandra**, Nadia L. Zakamska, et al. (2026)  
 “Double White Dwarf Binaries in SDSS-V DR19: A Catalog of DA White Dwarf Binaries and Constraints on the Binary Population”  
[\*The Astrophysical Journal\*, 1002, 207](#)
11. Madeline Lucey, **Vedant Chandra**, Alexander P. Ji, et al. (2026)  
 “Discovery of the First Five Carbon-enhanced Metal-poor Stars in the LMC”  
[\*The Astrophysical Journal Letters\*, 1000, L44](#)
10. Gautham A. Pallathadka\*, **Vedant Chandra**, Boris T. Gänsicke, et al. (2026)  
 “Double White Dwarf Binaries in SDSS-V DR19: The Discovery of a Rare DA+DQ White Dwarf Binary with a 31 hr Orbital Period”  
[\*The Astrophysical Journal\*, 999, 251](#)
9. Nicole R. Crumpler\*, **Vedant Chandra**, Nadia L. Zakamska, et al. (2025)  
 “A Large Catalog of DA White Dwarf Characteristics Using SDSS and Gaia Observations”  
[\*The Astrophysical Journal\*, 989, 24](#)
8. Christian Aganze, **Vedant Chandra**, Risa H. Wechsler, et al. (2025)  
 “The Cocytos Stream: A Disrupted Globular Cluster from our Last Major Merger?”  
[\*The Astrophysical Journal\*, submitted](#)
7. Dennis Zaritsky, **Vedant Chandra**, Charlie Conroy, et al. (2025)  
 “Untangling Magellanic Streams”  
[\*The Open Journal of Astrophysics\*, 8, 16](#)
6. Nicole R. Crumpler\*, **Vedant Chandra**, Nadia L. Zakamska, et al. (2024)  
 “Detection of the Temperature Dependence of the White Dwarf Mass–Radius Relation with Gravitational Redshifts”  
[\*The Astrophysical Journal\*, 977, 237](#)
5. Hans-Walter Rix, **Vedant Chandra**, Gail Zasowski, et al. (2024)  
 “The Extremely Metal-rich Knot of Stars at the Heart of the Galaxy”  
[\*The Astrophysical Journal\*, 975, 293](#)
4. Gautham A. Pallathadka\*, **Vedant Chandra**, Nadia L. Zakamska, et al. (2024)  
 “Discovery of a Proto–White Dwarf with a Massive Unseen Companion”  
[\*The Astrophysical Journal\*, 968, 42](#)
3. Stefan Arseneau\*, **Vedant Chandra**, Hsiang-Chih Hwang, et al. (2024)  
 “Measuring the Mass–Radius Relation of White Dwarfs Using Wide Binaries”  
[\*The Astrophysical Journal\*, 963, 17](#)
2. Hans-Walter Rix, **Vedant Chandra**, René Andrae, et al. (2022)  
 “The Poor Old Heart of the Milky Way”  
[\*The Astrophysical Journal\*, 941, 45](#)
1. Evan B. Bauer, **Vedant Chandra**, Ken J. Shen, J. J. Hermes (2021)  
 “Masses of White Dwarf Binary Companions to Type Ia Supernovae Measured from Runaway Velocities”  
[\*The Astrophysical Journal Letters\*, 923, L34](#)

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## Co-Authored Publications

40. Richard A. N. Brooks, Jason L. Sanders, **Vedant Chandra**, et al. (2026)  
 “The Milky Way–Large Magellanic Cloud interaction with simulation-based inference”  
[\*Monthly Notices of the Royal Astronomical Society\*, 550, in press](#)
39. Manuel Bayer, Else Starkenburg, Akshara Viswanathan, et al. (2026)  
 “Tracing the very early disruption of the Sagittarius dwarf galaxy in the distant Milky Way halo”  
[\*Monthly Notices of the Royal Astronomical Society\*, 550, in press](#)

38. Stefan Arseneau\*, J. J. Hermes, **Vedant Chandra**, et al. (2026)  
 “An Ultramassive White Dwarf with a Likely Oxygen–Neon Core”  
*The Astrophysical Journal*, 1006, 217
37. SDSS Collaboration, Mojgan Aghakhanloo, David Aguilar, et al. (2026)  
 “The Twentieth Data Release of the Sloan Digital Sky Survey: First All-Sky BOSS Spectra, eROSITA-SDSS-V Mapper Coordinated Observations, and a Preview of the Local Volume Mapper”  
*submitted*
36. Ilija Medan, Andrew R. Casey, Alexander P. Ji, et al. (2026)  
 “BOSS-CLAM: Utilizing a Constrained Linear Absorption Model to Infer Stellar Parameters from BOSS Spectra”  
*submitted*
35. Richard A. N. Brooks, Jason L. Sanders, Adam M. Dillamore, et al. (2026)  
 “Quantifying the Milky Way, LMC and their interaction using all-sky kinematics of outer halo stars”  
*Monthly Notices of the Royal Astronomical Society*, 549, *in press*
34. SDSS Collaboration, Mojgan Aghakhanloo, James Aird, et al. (2026)  
 “The Nineteenth Data Release of the Sloan Digital Sky Survey”  
*The Astrophysical Journal Supplement Series*, 285, 9
33. Kareem El-Badry, Klaus Werner, Ken J. Shen, et al. (2026)  
 “A systematic survey for hypervelocity runaways from thermonuclear supernovae”  
*The Open Journal of Astrophysics*, 9, 64326
32. Anya Phillips, Charlie Conroy, Jacob Nibauer, et al. (2026)  
 “The Binary Properties of Stellar Streams Are Set by Cluster Dynamics”  
*The Astrophysical Journal*, 1003, 188
31. Keyi Ding, Mario Gennaro, Roberto J. Avila, et al. (2026)  
 “Probing the IMF in the Early Universe – Direct measurements in the Boötes I UFD with JWST/NIRCam”  
*The Astrophysical Journal*, *in press*
30. Riley Thai\*, Andrew R. Casey, Alexander P. Ji, et al. (2026)  
 “Evaluating Classifications of Extremely Metal-poor Candidates Selected from Gaia XP Spectra”  
*The Astronomical Journal*, 171, 235
29. Roger E. Cohen, Mario Gennaro, Matteo Correnti, et al. (2026)  
 “The Stellar Initial Mass Function down To  $0.16 M_{\odot}$  toward the Small Magellanic Cloud”  
*The Astrophysical Journal*, 1000, 151
28. Juna A. Kollmeier, Hans-Walter Rix, Conny Aerts, et al. (2026)  
 “Sloan Digital Sky Survey. V. Pioneering Panoptic Spectroscopy”  
*The Astronomical Journal*, 171, 52
27. Cheyanne Shariat, Kareem El-Badry, Mario Gennaro, et al. (2025)  
 “Wide Binaries in an Ultra-faint Dwarf Galaxy: Discovery, Population Modeling, and A Nail in the Coffin of Primordial black hole Dark Matter”  
*Publications of the Astronomical Society of the Pacific*, 137, 104103
26. Stefan Arseneau\*, J. J. Hermes, Nadia L. Zakamska, et al. (2025)  
 “Resolution-corrected White Dwarf Gravitational Redshifts Validate Fifth-generation Sloan Digital Sky Survey Wavelength Calibration and Enable Accurate Mass–Radius Tests”  
*The Astrophysical Journal*, 991, 190
25. Pranav Nagarajan, Kareem El-Badry, Henrique Reggiani, et al. (2025)  
 “A Spectroscopic Search for Dormant Black Holes in Low-metallicity Binaries”  
*Publications of the Astronomical Society of the Pacific*, 137, 094202

24. Amanda Byström, Sergey E. Koposov, Sophia Lilleengen, et al. (2025)  
 “Exploring the interaction between the MW and LMC with a large sample of blue horizontal branch stars from the DESI survey”  
*Monthly Notices of the Royal Astronomical Society*, 542, 560
23. Angus Beane, James W. Johnson, Vadim A. Semenov, et al. (2025)  
 “Rising from the Ashes. II. The Bar-driven Abundance Bimodality of the Milky Way”  
*The Astrophysical Journal*, 985, 221
22. Gagik Tovmassian, Keith Inight, Anna Francesca Pala, et al. (2025)  
 “V498 Hya, a new candidate for a period bouncer cataclysmic variable”  
*Monthly Notices of the Royal Astronomical Society*, 537, 3234
21. Neige Frankel, Rene Andrae, Hans-Walter Rix, et al. (2025)  
 “What Does the LMC Look Like? It Depends on [M/H] and Age”  
*The Astrophysical Journal*, 979, 136
20. Turner Woody, Charlie Conroy, Phillip Cargile, et al. (2025)  
 “The Rapid Formation of the Metal-poor Milky Way”  
*The Astrophysical Journal*, 978, 152
19. Rohan P. Naidu, Jorjyt Matthee, Ivan Kramarenko, et al. (2024)  
 “All the Little Things in Abell 2744: >1000 Gravitationally Lensed Dwarf Galaxies at  $z = 0 - 9$  from JWST NIRCам Grism Spectroscopy”  
*The Open Journal of Astrophysics*, submitted
18. Vadim A. Semenov, Charlie Conroy, **Vedant Chandra**, et al. (2024)  
 “Formation of Galactic Disks. II. The Physical Drivers of Disk Spin-up”  
*The Astrophysical Journal*, 972, 73
17. Jiwon Jesse Han, Charlie Conroy, Dennis Zaritsky, et al. (2024)  
 “Our Halo of Ice and Fire: Strong Kinematic Asymmetries in the Galactic Halo”  
*The Astrophysical Journal*, submitted
16. Guilherme Limberg, Alexander P. Ji, Rohan P. Naidu, et al. (2024)  
 “Extending the chemical reach of the H3 survey: detailed abundances of the dwarf-galaxy stellar stream Wukong/LMS-1”  
*Monthly Notices of the Royal Astronomical Society*, 530, 2512
15. Jiadong Li, Kaze W. K. Wong, David W. Hogg, et al. (2024)  
 “AspGap: Augmented Stellar Parameters and Abundances for 37 Million Red Giant Branch Stars from Gaia XP Low-resolution Spectra”  
*The Astrophysical Journal Supplement Series*, 272, 2
14. Logan Sizemore, Diego Llanes, Marina Kounkel, et al. (2024)  
 “A Self-consistent Data-driven Model for Determining Stellar Parameters from Optical and Near-infrared Spectra”  
*The Astronomical Journal*, 167, 173
13. Vadim A. Semenov, Charlie Conroy, **Vedant Chandra**, et al. (2024)  
 “Formation of Galactic Disks. I. Why Did the Milky Way’s Disk Form Unusually Early?”  
*The Astrophysical Journal*, 962, 84
12. Alexander P. Ji, Sanjana Curtis, Nicholas Storm, et al. (2024)  
 “Spectacular Nucleosynthesis from Early Massive Stars”  
*The Astrophysical Journal Letters*, 961, L41
11. K. Inight, Boris T. Gänsicke, A. Schwöpe, et al. (2023)  
 “Cataclysmic Variables from Sloan Digital Sky Survey - V. The search for period bouncers continues”  
*Monthly Notices of the Royal Astronomical Society*, 525, 3597

10. Andrés Almeida, Scott F. Anderson, Maria Argudo-Fernández, et al. (2023)  
 “The Eighteenth Data Release of the Sloan Digital Sky Surveys: Targeting and First Spectra from SDSS-V”  
[\*The Astrophysical Journal Supplement Series\*, 267, 44](#)
9. Kareem El-Badry, Ken J. Shen, **Vedant Chandra**, et al. (2023)  
 “The fastest stars in the Galaxy”  
[\*The Open Journal of Astrophysics\*, 6, 28](#)
8. René Andrae, Hans-Walter Rix, **Vedant Chandra** (2023)  
 “Robust Data-driven Metallicities for 175 Million Stars from Gaia XP Spectra”  
[\*The Astrophysical Journal Supplement Series\*, 267, 8](#)
7. Jiwon Jesse Han, Charlie Conroy, Benjamin D. Johnson, et al. (2022)  
 “The Stellar Halo of the Galaxy is Tilted and Doubly Broken”  
[\*The Astronomical Journal\*, 164, 249](#)
6. Jiwon Jesse Han, Rohan P. Naidu, Charlie Conroy, et al. (2022)  
 “A Tilt in the Dark Matter Halo of the Galaxy”  
[\*The Astrophysical Journal\*, 934, 14](#)
5. Hsiang-Chih Hwang, Yuan-Sen Ting, Charlie Conroy, et al. (2022)  
 “Wide binaries from the H3 survey: the thick disc and halo have similar wide binary fractions”  
[\*Monthly Notices of the Royal Astronomical Society\*, 513, 754](#)
4. Rohan P. Naidu, Charlie Conroy, Ana Bonaca, et al. (2022)  
 “Live Fast, Die  $\alpha$ -Enhanced: The Mass-Metallicity- $\alpha$  Relation of the Milky Way’s Disrupted Dwarf Galaxies”  
[\*The Astrophysical Journal\*, submitted](#)
3. Charlie Conroy, David H. Weinberg, Rohan P. Naidu, et al. (2022)  
 “Birth of the Galactic Disk Revealed by the H3 Survey”  
[\*The Open Journal of Astrophysics\*, submitted](#)
2. Rohan P. Naidu, Alexander P. Ji, Charlie Conroy, et al. (2022)  
 “Evidence from Disrupted Halo Dwarfs that r-process Enrichment via Neutron Star Mergers is Delayed by  $\sim 500$  Myr”  
[\*The Astrophysical Journal Letters\*, 926, L36](#)
1. Evan Petrosky, Hsiang-Chih Hwang, Nadia L. Zakamska, et al. (2021)  
 “Variability, periodicity, and contact binaries in WISE”  
[\*Monthly Notices of the Royal Astronomical Society\*, 503, 3975](#)

## Unrefereed Publications

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6. **Via Collaboration** (2026)  
 “The Via Project: Overview of the Science, Instrument, and Survey”  
[\*White Paper\*](#)
5. Sy Boles & **Vedant Chandra** (2025)  
 “Strange galactic facts”  
[\*The Harvard Gazette\*](#)
4. Arjun Dey, Joan Najita, Carrie Fillion, et al (2023)  
 “RomAndromeda: The Roman Survey of the Andromeda Halo”  
*NASA Roman Core Community Survey White Paper*
3. Jiwon Jesse Han, Arjun Dey, Adrian M. Price-Whelan, et al (2022)  
 “NANCY: Next-generation All-sky Near-infrared Community surveyY”  
*NASA Roman Core Community Survey White Paper*

2. Charlie Conroy, Dan Fabricant, Nelson Caldwell, **Vedant Chandra**, et al (2022)  
“A Fast All-Sky Spectroscopic Survey to Discover the Nature of Dark Matter, Find the Edge of Galaxy Formation, and Map the Cold Gas Feeding the Milky Way”  
*CfA Science & Technology White Paper*
1. **Vedant Chandra** (2020)  
“Measuring the White Dwarf Mass-Radius Relation using Thousands of Stars”  
[astrobites](#)